NLP Project: Topic Modeling & LDA Visualization

Advanced Natural Language Processing Pipeline

1. Project Overview

Is project mein humne **Latent Dirichlet Allocation (LDA)** technique use karte hue text data mein se hidden topics extract kiye hain. Iske saath humne ek interactive **LDA Visualization** generate ki hai jo topics ke darmiyan relationship aur unke keywords ko display karti hai.

2. Technical Workflow

Step	Description
Preprocessing	Tokenization, Stopwords removal, aur Lemmatization (using NLTK/Gensim).
Dictionary Creation	Text ko unique IDs mein convert karna (Bag of Words).
LDA Modeling	Unsupervised learning algorithm ke zariye statistical topics identify karna.
Visualization	**pyLDAvis** library ka use kar ke interactive clusters aur inter-topic distance map banana.

3. Topic Modeling Concept (LDA)

LDA ka kaam ye hai ke wo text mein se un words ko group karta hai jo aksar ek saath aate hain. Har group ek "Topic" kehlata hai. Maslan, agar keywords "model", "flask", aur "app" hain, toh ye **Web Development** ka topic ho sakta hai.

4. Implementation Highlights

```
# LDA Model Training Snippet lda_model = gensim.models.ldamodel.LdaModel( corpus=corpus, id2word=id2word, num_topics=5, passes=10, alpha='auto' ) # Interative Visualization vis = pyLDAvis.gensim.prepare(lda_model, corpus, id2word) pyLDAvis.save_html(vis, 'lda_visualization.html')
```

5. Key Features

- **Interactive Bubbles:** Har bubble ek topic ko represent karta hai. Bubble jitna bada hoga, topic utna dominant hoga.
- **Saliency:** Words ko unki frequency aur uniqueness ke mutabiq rank kiya jata hai.
- **Inter-topic Distance:** Ye dikhata hai ke topics aapas mein kitne milte julte hain.

6. Conclusion

Ye project massive amounts of unlabelled text data ko summarize karne ke liye behtareen hai. Is ke results humein hidden insights dhoondne mein madad dete hain jo manually parhna mumkin nahi hota.